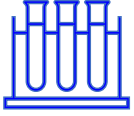


Common Scenario

Click these buttons to
REVEAL & HIDE answers
and graphics

Step 1



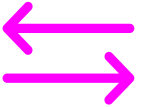
Step 2



Step 3



Step 4



Step 5

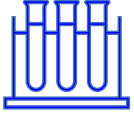


Cases

Inpatient Management of Hyperkalemia

Common Scenario

Step 1



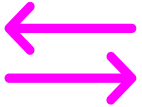
Step 2



Step 3



Step 4



Step 5



Cases

You are on a hospital wards rotation pre-rounding for the day when you get a call from the nurse for Mrs. Smith...

Nurse: "I have a critical lab to report on Mrs. Smith"

You: "What is the critical lab?"

Nurse: "The patient's potassium is 6.7..."

Question: What do you do next?

Common Scenario

Step 1



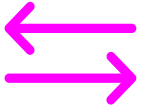
Step 2



Step 3



Step 4



Step 5



Cases

$K > 5.5?$

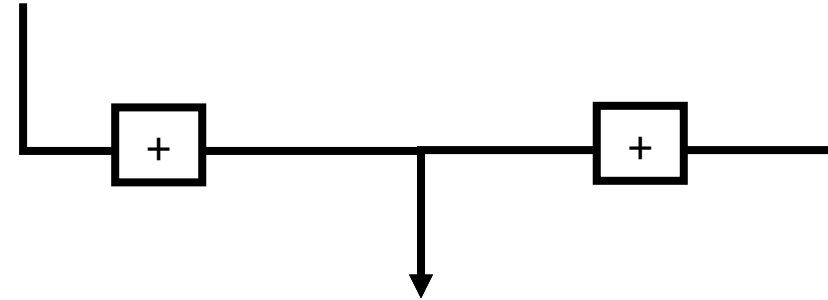
Confirm hyperK

Considerations



Is this result unexpected?

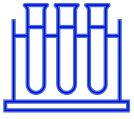
Is there hemolysis?



Repeat potassium

Common Scenario

Step 1



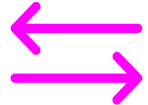
Step 2



Step 3



Step 4



Step 5



Cases

K > 5.5?

Confirm hyperK

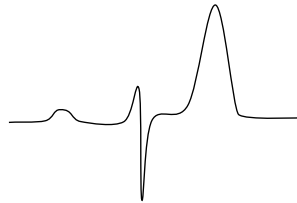
Stat ECG

Obtain STAT ECG

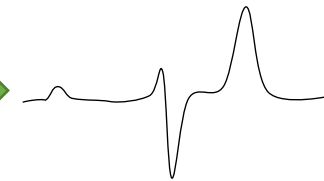
Hyperkalemia can cause life-threatening arrhythmias (e.g., sinus bradycardia, sinus arrest, asystole, VT/VF)

Obtain ECG for any K > 5.5

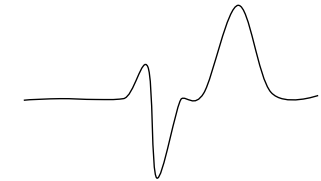
Look for characteristic, progressive ECG changes



Peaked T waves
QT shortening



PR prolongation
QRS widening

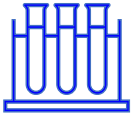


Loss of P waves
QRS widens further

Repeat ECG every 30 – 60 min to ensure resolution of abnormalities
Monitor on telemetry

Common Scenario

Step 1



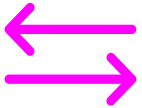
Step 2



Step 3



Step 4



Step 5



Cases

K > 5.5?

Confirm hyperK

Stat ECG

ECG Δ /
K > 6.5

IV Ca
f/u ECG

Cardiac Membrane Stabilization

ECG changes or K > 6.5

Calcium Chloride 0.5-1 g IV

Calcium Gluconate 1-2 g IV

More potent, requires central access

Can be given peripherally

Onset: 3-5 minutes

Duration: 30-60 minutes

No effect on K⁺ levels but should normalize ECG changes

Repeat ECG

Redose IV Ca every 5-10 min if
persistent ECG changes

Common Scenario

Step 1



Step 2



Step 3



Step 4



Step 5



Cases

K > 5.5?

Confirm hyperK

Stat ECG

ECG Δ /
K > 6.5

IV Ca
f/u ECG

Insulin+D50W
Albuterol
±NaHCO₃

Potassium Shifting

Shifting K intracellularly is a **temporizing measure** to buy time for more definitive management

Insulin + D50

IV 10U* Regular Insulin
+ 1 amp D50

*IV 5U Regular insulin if
renal impairment

↓ K ~0.5-1.0 mEq

B2 agonists

Albuterol 10-20mg in
4mL saline
Higher than asthma/COPD
dosing (2.5mg)

↓ K 0.5-1.5 mEq

Na HCO₃

50-100 mEq IV

For non-ESRD with
acidosis only

Least effective shifting
strategy

1 h

2 h

3 h

4 h

5 h

6 h

Insulin + D50

Albuterol

HCO₃

If
AKI

Common Scenario

Step 1



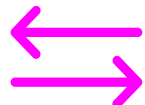
Step 2



Step 3



Step 4



Step 5



Cases

K > 5.5?

Confirm hyperK

Stat ECG

ECG Δ /
K > 6.5

IV Ca
f/u ECG

Insulin+D50W
Albuterol
±NaHCO₃

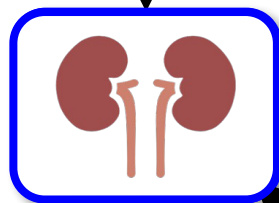
IVF ± loops
HD
Lokelma
Kayexalate

Repeat K
in 1-4 h

Potassium Elimination

90%

10%



↑ distal Na
delivery

K cleared in
dialysate

IVF

Loop diuretic

HD/ CRRT

Lokelma

Kayexalate

1st line if ↓ V
+ loops if
euvolemic

IV 40-160 mg
1st line if ↑ V
+ IVF if euvolemic

Used in acute or
chronic renal
failure

\$\$\$
Large Na load
Limited

Risk of bowel
necrosis (SBO,
post-op)

30 min

1 h

2 h

3 h

4 h

5 h

6 h

12 h

24 h

48 h

NS, NaHCO₃

Loops

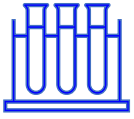
HD/CRRT

Lokelma 10 g TID

Kayexalate 15-60 g q6h

Common Scenario

Step 1



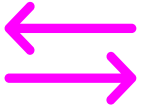
Step 2



Step 3



Step 4



Step 5



Cases

Case 1

55 yo with ESRD on HD via an AV fistula presents with shortness of breath and pain in their fistula site after missing dialysis. Fistulogram shows stenosis of the AV fistula. They are anuric.

BMP - K 6.7 (no hemolysis), HCO₃ 19, BUN 56, Cr of 10.5.

What is the next step in management?

Obtain an ECG.



What is the next step in management?

ECG – no obvious changes. With the severity of hyperkalemia, give **IV Ca** (gluconate 1- 2 g) for cardiac stabilization. Monitor on telemetry.

What temporizing measures would you use?

IV regular **insulin 5U** (because of ESRD) + D50W, **albuterol** nebulizer 20 mg. NaHCO₃ likely ineffective in ESRD.

What medications will you use for K elimination?

SZC if available, **Kayexalate** and urgent reversal of fistulogram. Loop diuretics are ineffective when anuria.

Case 2

65 yo with metastatic colon cancer presents for colectomy. On POD3, develops new non-oliguric AKI.

Appears euvolemic on exam. Labs show K of 6.0, Cr 0.6 → 3.5, HCO₃ 13. ABG shows pH of 7.2.

What is the next step in management?

Obtain an ECG. Click below for full image.



What is the next step in management?

IV Ca gluconate 1-2 g. **Repeat ECG** in 30-60 min. Repeat IV Ca if persistent ECG changes. Monitor on telemetry.

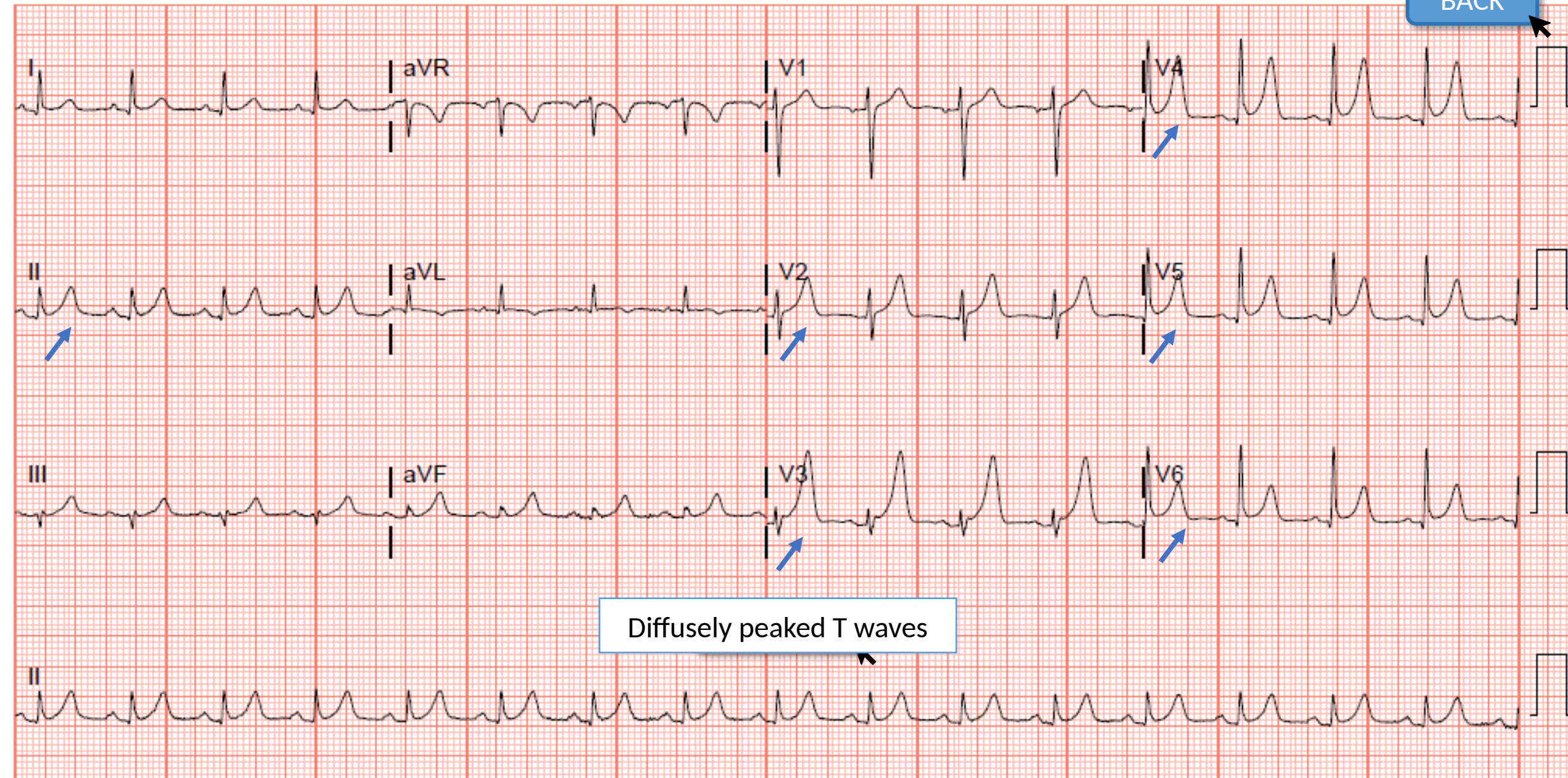
What temporizing measures would you use?

IV regular **insulin 5U** (because of AKI) + D50W, **albuterol** nebulizer, **NaHCO₃** given acidosis

What medications will you use for K elimination?

IV **loop diuretics** +/- **IVF** (isotonic NaHCO₃ infusion) to avoid worsening Cr. No Kayexalate given recent bowel surgery

BACK



Guideline Update: UKKA 2023 & KDIGO 2024 (original teaching unchanged)

UK Kidney Association — Management of Hyperkalaemia in Adults (Oct 2023)

Emergency thresholds match this talk: IV calcium for ECG changes; treat $K \geq 6.5$ mmol/L as an emergency.

Insulin-glucose + nebulized β 2-agonist remain first-line shifting; monitor glucose ≥ 6 h after insulin (hypoglycemia is the most common complication).

NEW: sodium zirconium cyclosilicate (SZC/Lokelma) 10 g TID up to 72 h is now an option ALONGSIDE shifting agents in acute life-threatening hyperkalemia ($K \geq 6.5$).

Sodium polystyrene sulfonate (Kayexalate) is NOT recommended for emergency treatment — reserve for non-urgent settings when SZC/patiromer unavailable.

KDIGO 2024 CKD Guideline

RASi-associated hyperkalemia: lower the potassium (diuretics, binders, dietary review, bicarbonate if acidotic) rather than reflexively stopping the ACEi/ARB — RASi discontinuation is linked to higher mortality and CV events in CKD.

Sources: UKKA Clinical Practice Guideline, Management of Hyperkalaemia in Adults (2023); KDIGO 2024 CKD Guideline (Kidney Int 2024).

Take-Home Points

1. Hyperkalemic crisis is a medical emergency — severe weakness and fatal arrhythmias. Any $K > 5.5$ → STAT ECG; treat $K > 6.5$ as emergent regardless of ECG.
2. ECG changes: peaked T waves → P-wave loss → QRS widening → sine wave/VF. ECG changes or $K > 6.5$ → IV calcium gluconate 1–2 g (chloride 0.5–1 g via central line); re-dose in 5–10 min if ECG unchanged.
3. Shift K intracellularly while you arrange elimination: regular insulin 10U IV + D50 (5U if renal impairment), albuterol neb 10–20 mg, $\pm$ NaHCO₃ (only if metabolic acidosis, not ESRD).
4. Eliminate K: loop diuretics $\pm$ isotonic fluids if volume allows; binders (SZC 10 g TID preferred over Kayexalate) if renal dysfunction; urgent dialysis for severe renal failure or refractory hyperkalemia.
5. Recheck K in 1–4 h and repeat ECG every 30–60 min until abnormalities resolve; shifting only buys time — it does not remove potassium.

Attribution & Changelog

Original

"Inpatient Management of Hyperkalemia" — TeachIM.org (Dec 2021). Authors: Tiffany Gardner MD, Caitlin Dietsche MD, Yilin Zhang MD. Exec. editor: Yilin Zhang MD. https://teachim.org/teaching_material/hyperkalemia-3/ — CC BY-NC 4.0 (educational, non-commercial use).

Corrections in this optimized copy (June 2026)

Slide 7: Lokelma "10 mg TID" → "10 g TID"; Kayexalate "15-60 mg q6h" → "15-60 g q6h" (unit errors; doses per lesson text & UKKA 2023).

Slide 8 Case 1: albuterol "nebulizer 20 g" → "20 mg" (unit error; deck slide 6 & lesson text say 10–20 mg).

Spelling: "Kayexelate" → "Kayexalate" (8×, slides 7–8).

Appended slides only — all original 9 slides, animations, and click-to-reveal builds preserved.

Companion files

Facilitator-guide-1.docx · hyperkalemia-worksheet.docx · Hyperkalemia_Lesson_Package.html · Hyperkalemia_OnePager.pdf · Hyperkalemia_Anki_Cards.txt